ODE - 3

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First order ODEs

① Separable variables

$$dy/dx = g(x)h(y)$$

$$\Rightarrow \int \frac{dy}{h(y)} dy = \int g(x)dx + c$$

Example

$$3e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$$

$$\Rightarrow 3e^x \tan y\, dx = (e^x - 1)\sec^2 y\, dy$$

$$\Rightarrow \frac{3e^x}{e^x - 1} dx = \frac{\sec^2 y}{\tan y} dy$$

Integrating,

$$3\ln(e^x - 1) = \ln(c' \tan y)$$

$$\Rightarrow (e^x - 1)^3 = c' \tan y$$

② Linear first-order ODE

$$dy/dx + P(x)y = Q(x)$$

Integrating factor: $IF = e^{\int P(x)dx}$

General soln: $y \cdot IF = \int Q(x) \cdot IF\, dx + c$

Example

$$dy/dx + 2y = x \Rightarrow P(x) = 2,\ Q(x) = x$$

① $IF = e^{\int 2dx} = e^{2x}$

IF is multiplied on both sides,

$$\frac{d}{dx}(e^{2x} \cdot y) = x \cdot e^{2x}$$

Integrating, $e^{2x} \cdot y = \frac{x^2}{2} + c$ (arbit. const.)

Solve

② $dy/dx + \frac{2y}{x} = \sin x$

Integrating factor $= e^{\int \frac{2dx}{x}} = e^{2\ln x} = x^2$

IF is multiplied on both sides,

$$\frac{dy}{dx} \cdot x^2 + 2xy = x^2 \sin x$$

$$\frac{d}{dx}(x^2 y) = x^2 \sin x$$

Integrating,

$$x^2 y = \int x^2 \sin x + c$$

$$= -x^2 \cos x + 2x \sin x + 2\cos x + c$$

(Integration by parts)

③ $x \cos x \frac{dy}{dx} + y(\sin x + \cos x) = 1$

We write the equation in standard form

$$\frac{dy}{dx} + \frac{x \sin x + \cos x}{x \cos x} y = \frac{1}{x \cos x} \quad \text{—(1)}$$

$$I.F = e^{\int P dx} = e^{\int \frac{x \sin x + \cos x}{x \cos x} dx}$$

$$= e^{\int \tan x\, dx + \int \frac{dx}{x}}$$

$$= e^{-\ln(\cos x) + \ln(x)}$$

$$= e^{\ln(\sec x) + \ln x}$$

$$= x \sec x$$

I.F × (1) ⇒

$$x \sec x \cdot \frac{dy}{dx} + \frac{x \sin x + \cos x}{\cos^2 x} y = \frac{1}{\cos^2 x}$$

$$\Rightarrow x \sec x \cdot \frac{dy}{dx} + [x \sec x \tan x + \sec x] y = \sec^2 x$$

$$\therefore \frac{d}{dx}[y \cdot x \sec x] = \sec^2 x$$

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homogeneous → $\frac{dy}{dx} + P(x) \cdot y = 0$

$$\Rightarrow \frac{dy}{dx} = -y \cdot P(x)$$

$$\Rightarrow \int \frac{dy}{y} = -\int P(x)dx + c$$

nonhomogeneous → $\frac{dy}{dx} + P(x) \cdot y = Q(x)$

$$\frac{dy}{dx} + 2y = x \quad \text{—(1)}$$

$\frac{dy}{dx} + P(x) \cdot y = Q(x)$ (nonhomogeneous ODE)

$\Rightarrow P(x) = 2,\ Q(x) = x$

Integrating factor (IF) $= e^{\int P(x)dx}$

$= e^{\int 2dx} = e^{2x}$

Multiplying both sides of (1) by e^{2x},

$$e^{2x} \cdot \frac{dy}{dx} + 2y \cdot e^{2x} = x \cdot e^{2x}$$

$$\Rightarrow \frac{d}{dx}(y \cdot e^{2x}) = x \cdot e^{2x}$$

$$\Rightarrow d(y \cdot e^{2x}) = x \cdot e^{2x} dx$$

Integrating, $y \cdot e^{2x} = \int x \cdot e^{2x} dx + c$

$$= \frac{x \cdot e^{2x}}{2} - \int \frac{e^{2x}}{2} \cdot dx + c$$

$$= \frac{x \cdot e^{2x}}{2} - \frac{1}{4} \cdot e^{2x} + c$$

$\Rightarrow x\sec x \cdot \frac{dy}{dx} + [illegible]$

$\Rightarrow \frac{d}{dx}[y \cdot x\sec x] = \sec^2 x$

$\Rightarrow d(y \cdot x\sec x) = \sec^2 x\,dx$

$\Rightarrow$ Integrating,

$y x \sec x = \tan x + c$

$xy = \sin x + c\cos x$

(4) $y^2 + (x - \frac{1}{y})\frac{dy}{dx} = 0$

can we put this into this form? Let us try

$y^3 + (xy - 1)\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} + \frac{y^3}{xy-1} = 0$

$\frac{dy}{dx} + P(x)y = Q(x) \Rightarrow dx/dy + P(y)x = Q(y)$

Here independent variable is y

$\Rightarrow \frac{dx}{dy} \cdot y^3 + (xy - 1) = 0$

$\Rightarrow \frac{dx}{dy} + \frac{x}{y^2} = \frac{1}{y^3}$

$IF = e^{\int P\,dy} = e^{\int y^{-2} dy}$

$= e^{-1/y}$

$\Rightarrow e^{-1/y}\left(\frac{dx}{dy} + \frac{x}{y^2}\right) = \frac{1}{y^3} \cdot e^{-1/y}$

$\Rightarrow \frac{d}{dy}\left(x \cdot e^{-1/y}\right) = \frac{1}{y^3} \cdot e^{-1/y}$

$\Rightarrow d(x \cdot e^{-1/y}) = \frac{1}{y^3} \cdot e^{-1/y}\,dy$

Integrating,

$x \cdot e^{-1/y} = e^{-1/y}\left(1 + \frac{1}{y}\right) + c$

is the required general soln.

(5) $\frac{dy}{dx} + 2xy = x$

$IF = e^{\int 2x\,dx} = e^{x^2}$

$y \cdot e^{x^2} = \int x e^{x^2} dx + c$

↳ substitute $u = x^2$, $\frac{du}{dx} = 2x$

$= \int \frac{e^u}{2} \cdot du + c$

$= \frac{1}{2} \cdot e^u + c$

$= \frac{1}{2} \cdot e^{x^2} + c$

⑥ $\frac{dy}{dx} - \frac{3y}{x} = x^2$

$IF = e^{-\int \frac{3dx}{x}} = e^{-3\ln x} = \frac{1}{x^3}$

$\Rightarrow \frac{d}{dx}\left[y \cdot \frac{1}{x^3}\right] = \frac{1}{x}$

$\Rightarrow \boxed{yx^{-3} = \ln|x| + C_1}$

⑦ Try: $\frac{dy}{dx} + by = ae^{-\lambda x} \quad (\lambda > 0)$